

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 7: Eigensystems

Adam Prugel-Bennett

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1 Keywords

- Eigenvectors, Orthogonal Matrices, Eigenvector Decomposition

2 Main Points

2.1 Eigen-Systems

- We can understand ill-conditioning for linear regression by the eigen-decomposition of $\mathbf{M} = \mathbf{X}^T \mathbf{X}$
- This should be revision
- You know that an *eigenvector*, $\mathbf{v}$, satisfies $\mathbf{M} \mathbf{v} = \lambda \mathbf{v}$
- For a symmetric matrix there are n real orthogonal eigenvectors
- You can prove they are orthogonal
- **Orthogonal Matrices**
 - Putting the n eigenvectors into a matrix $\mathbf{V}$ with columns $\mathbf{v}_i$ we obtain an orthogonal matrix
 - The defining property of an orthogonal matrix is $\mathbf{V}^T \mathbf{V} = \mathbf{V} \mathbf{V}^T = \mathbf{I}$
 - They correspond to rotations (with a possible reflection)
- **Matrix Decomposition**
 - We can decompose a symmetric matrix, $\mathbf{M}$ as
$$\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$$
 - Where $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues of $\mathbf{M}$ (i.e. $\Lambda_{ii} = \lambda_i$)
 - $\mathbf{V}$ is the orthogonal matrix made up of the eigenvectors of $\mathbf{M}$
 - We can interpret the mapping of a symmetric matrix $\mathbf{M}$ as equivalent to
 1. a rotation defined by $\mathbf{V}^T$
 2. scaling of the i^{th} component by λ_i and
 3. a rotation backwards given by $\mathbf{V}$
 - Equivalently if we work in a coordinate system defined by the eigenvectors of $\mathbf{M}$ (this forms an orthonormal basis set) then we just rescale in the directions $\mathbf{v}_i$ by λ_i
 - A symmetric matrix just squashes or expands in different orthogonal directions (this is what the eigensystem captures)

- **Inverse Matrices**

- The inverse of a symmetric matrix $\mathbf{M}$ is given by

$$\mathbf{M}^{-1} = \mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^T$$

- Where $\mathbf{\Lambda}^{-1}$ is a diagonal matrix with elements $\Lambda_{ii}^{-1} = 1/\lambda_i$
- This is only defined if all the eigenvalues of $\mathbf{M}$ are non-zero ($\mathbf{M}$ is said to be full rank)
- If λ_i is very small then $1/\lambda_i$ is large and in taking the inverse $\mathbf{M}^{-1}\mathbf{x}$ any component of $\mathbf{x}$ in the direction $\mathbf{v}_i$ will get magnified by $1/\lambda_i$
- For linear regression we invert $\mathbf{M} = \mathbf{X}^T \mathbf{X}$
 - in directions where the training examples don't vary much the associated eigenvalue will be small and the inverse inherently unstable

3 Exercises

1. Let $\mathbf{M}$ be a symmetric matrix with eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ and distinct eigenvalues $\lambda_1 \neq \lambda_2$. By computing $\mathbf{v}_2^T \mathbf{M} \mathbf{v}_1$ in two different ways, show that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal
2. Show that the eigenvalues of $\mathbf{X}^T \mathbf{X}$ are never negative. (Hint: consider $\|\mathbf{X} \mathbf{v}\|^2$)
3. Using $\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$
 - (a) Show that $\mathbf{M}^{-1} = \mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^T$ by multiplying the two together
 - (b) Show that $\mathbf{M}^k = \mathbf{V} \mathbf{\Lambda}^k \mathbf{V}^T$
4. Find the eigenvalues and eigenvectors of $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and check that $\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$

4 Experiments

4.1 Eigensystems

- In either Matlab/Octave or python generate random matrices and check the matrix identities

```
X = randn(5,4) % generate a mock design matrix with 5 inputs of length 4
M = X'*X       % compute a symmetric matrix
[V,L] = eig(M) % compute eigenvectors and eigenvalues
V*L*V'        % should be identical to M
V*V'          % should be the identity matrix (up to rounding precision)
V'*V          % should be the identity matrix (up to rounding precision)
x = randn(4,1) % generate a random column matrix of length 4
y = randn(4,1) % generate another random column matrix of length 4
xp = V*x       % apply V to x
yp = V*y       % apply V to y
norm(x)        % compute Euclidean norm of x
norm(xp)       % should be the same as the Euclidean norm of x
x'*y           % compute inner product of x and y
xp'*yp         % inner product of xp and yp (should be the same as above)

Z = rand(4,5)  % a design matrix with more unknowns than examples
W = Z'*Z      % a covariance-like matrix (but without subtracting the mean)
eig(W)        % compute eigenvalues (one should be 0 up to machine precision)
```

5 Answers

1. On one hand $\mathbf{v}_2^\top(\mathbf{M} \mathbf{v}_1) = \lambda_1 \mathbf{v}_2^\top \mathbf{v}_1$. On the other, as $\mathbf{M}^\top = \mathbf{M}$, $\mathbf{v}_2^\top \mathbf{M} \mathbf{v}_1 = (\mathbf{M} \mathbf{v}_2)^\top \mathbf{v}_1 = \lambda_2 \mathbf{v}_2^\top \mathbf{v}_1$. Subtracting, $(\lambda_1 - \lambda_2) \mathbf{v}_2^\top \mathbf{v}_1 = 0$ and since $\lambda_1 \neq \lambda_2$ we must have $\mathbf{v}_2^\top \mathbf{v}_1 = 0$
2. If $\mathbf{X}^\top \mathbf{X} \mathbf{v} = \lambda \mathbf{v}$ then $\lambda \|\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 \geq 0$, so $\lambda \geq 0$. It is zero exactly when $\mathbf{X} \mathbf{v} = \mathbf{0}$: a direction the data say nothing about
3. Matrix powers
 - (a) $\mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top \mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^\top = \mathbf{V} \mathbf{\Lambda} \mathbf{\Lambda}^{-1} \mathbf{V}^\top = \mathbf{V} \mathbf{V}^\top = \mathbf{I}$, using $\mathbf{V}^\top \mathbf{V} = \mathbf{I}$ twice
 - (b) In $\mathbf{M}^k = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top \dots \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top$ every inner $\mathbf{V}^\top \mathbf{V}$ is $\mathbf{I}$, leaving $\mathbf{V} \mathbf{\Lambda}^k \mathbf{V}^\top$
4. $\det(\mathbf{M} - \lambda \mathbf{I}) = (2 - \lambda)^2 - 1 = 0$ gives $\lambda = 3$ with $\mathbf{v} = (1, 1)^\top / \sqrt{2}$ and $\lambda = 1$ with $\mathbf{v} = (1, -1)^\top / \sqrt{2}$. Then

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} = \mathbf{M}$$